YASH LAHOTI

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EDUCATION

ICAHN SCHOOL OF MEDICINE, Mount Sinai — **New York, NY**

Doctor of Medicine Candidate (MD) - M4 — ***2022 - 2027***

UNIVERSITY OF PENNSYLVANIA, School of Engineering & Applied Sciences — **Philadelphia, PA**

Masters of Science in Engineering: Electrical - Artificial Intelligence & Machine Learning — ***2021***

Bachelors of Applied Science: Biomedical Science — ***2017 - 2021***

CLINICAL/ACADEMIC EXPERIENCE

NEW YORK EYE AND EAR INFIRMARY — **New York , NY**

Clinical Research Fellowship *- Dr. Alon Harris Lab* — ***2025 - Present***

- Developed a high-fidelity 10TB ophthalmic database by unifying raw imaging (OCT, OCTA), systemic vascular biomarkers, and longitudinal visual field (VF) perimetry from 5+ NIH studies to enable multimodal glaucoma progression analysis.
- Engineered an analytics pipeline for clinical validation of novel Doppler holography, developing advanced statistical methodologies and computer vision models to characterize hemodynamic biomarker differences in glaucoma patients.
- Validated multimodal deep learning frameworks to identify fast progressors in glaucoma; synthesized structural and functional data to provide clinicians with predictive risk-stratification for earlier surgical or medical intervention from EHR records.
- Directed the translational interface between clinical ophthalmologists and engineering teams, authoring technical grant proposals for novel diagnostic tools and managing the creation of AI-ready datasets into the hospital's research infrastructure.

Ocular Imaging Technician — ***2025 - Present***

- Performed ocular imaging for 100+ patients enrolled in clinical trials at NYEE, utilizing OCT, OCTA, Laser Doppler Holography, OcuMet, TearFilm Imager in healthy and glaucoma patients at the optic nerve head and macula.

ICAHN SCHOOL OF MEDICINE — **New York, NY**

Oculomics Translational Investigator - Barry Family Center for Ophthalmic AI & Human Health — ***2026 - Present***

- Architected plan for Sinai/NYEE ophthalmology research repository by harmonizing imaging, longitudinal EHR, and multi-omics data into AI-ready pipelines (OMOP-standardized; HPC-integrated; +2 Petabyte), enabling foundation model development for systemic disease detection from ocular biomarkers and supporting institutional research data access.
- Designed and led structured AI education initiatives for ophthalmology faculty and residents, translating deep learning, multimodal EHR integration, and imaging analytics (OCT/OCTA/HVF/fundus) into clinically actionable frameworks; building cross-departmental literacy in AI-driven study design and developing reproducible research pipelines.

Course Director, Artificial Intelligence in Medicine — ***2019 - 2020***

- Administered and taught the first iteration of AI education for 50+ medical students, developing a curriculum of 8 interactive coding workshops and F500 industry leader speaker-series lectures to educate how AI can augment clinical decision-making.

Admissions Committee Member — ***2025 - 2026***

CHO/KIM AI SPINE LAB, Mount Sinai — **New York, NY**

Lead AI Student Research Director — ***2022 - 2024***

- Directed engineering and experimental design of 15+ active AI initiatives, delegating responsibilities across 25 graduate students and development of 3 software prototypes into clinician workflows.
- Implemented RAG LLM-Agent System (PydanticAI) with 90% accuracy in assigning appropriate CPT billing codes to over 160K orthopedic operative notes and modify/augment medical documentation for successful insurance approvals.
- Established a proprietary longitudinal database of 1,200+ scoliosis cases, radiographic annotations and clinical outcomes to train AI models and evaluate against 10,000+ active patients across Sinai health system.

TECHNICAL EXPERIENCE

THETA TECH AI - *Healthcare Consultancy* — **New York, NY**

Ophthalmology & AI Consultant — ***2022 - Present***

- Directed prototyping and implementation for a multispectral tear film imaging device needing measurement automation solution; provided technical and clinical guidance for 6 engineers, assigning tasks, troubleshooting errors, and designing KPIs.

SPINESIGHT - *Targeted Healthcare Innovation Fellowship* — **New York, NY**

Co-Founder & Lead Engineer — ***2022 - Present***

- Spearheaded the technical development of SpineSight, an AI-driven platform that enhances quantification and diagnosis of spinal cord diseases, automating MRI measurements and reducing patient diagnostic time by 85% (12 months).
- Coordinated a multidisciplinary team of neurosurgeons and neuroradiologists to co-develop 140+ novel clinical metrics, bringing standardization to spinal disease tracking and enabling faster, personalized treatment pathways for patients.
- Conducted market research through 150+ interviews with clinicians and patients to refine point-of-access in clinical workflow.

TDK SENSEI - *Health Monitoring Solution Startup* — **Pittsburgh, PA**

Machine Learning Engineer — ***2022***

- Constructed a TinyML solution for predicting invasive cardiac pressures using non-invasive ECG data, based on over 2000+ hours of patient sensor data, achieving 95% prediction accuracy for central venous pressure in an outpatient facility.

PUBLICATIONS

Manuscripts:

- **Lahoti Y.**, Harris A., Siesky B., Kwan M., Rosen R. "Absolute quantification of retinal hemodynamics via laser Doppler holography: distinguishing disease-specific velocity deficits from pressure-modulated flow impairment in glaucoma". Br J Ophthalmol. (Under Review).
- Joyce M.*, Rai R.*, Guidoboni G., Keller J., Wikle C., **Lahoti Y.**, et al.. "Transfer learning methods from clinical research to population-based studies in glaucoma". Br J Ophthalmol. (Under Review).
- **Lahoti, Y**., Sai, S., Ahmed, W., Rajjoub, R., Li, M., Zaidat, B., et al. Development of a novel machine learning model to automate vertebral column segmentation utilizing biplanar full-body imaging. The Spine Journal 2025 (Accepted).
- Zaidat, B., **Lahoti, Y.**, Yu, A., et al. "Artificially Intelligent Billing in Spine Surgery: An Analysis of a Large Language Model." Sage 2023 (Accepted).
- Yu, J., **Lahoti, Y. S.**, McCandless, K. C., et al. "Automated Scoliosis Cobb Angle Classification in Biplanar Radiograph Imaging with Explainable Machine Learning Models." Spine Journal 2025 (Accepted).

Invited Lectureship:

- **"**Integrating Computational Modeling and Artificial Intelligence for Improved Diagnosis and Longitudinal Follow-Up in Ophthalmology" - 2026 SIAM Conference on the Life Sciences | Computational Modeling Approaches in Ocular Disease.

Abstracts & Posters:

- **Lahoti, Y.**, Guidoboni, G., Greenfield, J., Cohen, G. J., Potash, S., Siesky, B., Verticchio Vercellin, A., Wood, S. K., Kwan, M., Harris, A. "Automated capillary-level optical coherence tomography angiography phenotyping in patients with glaucoma and diabetes." SAIVO 2026 (Submitted).
- **Lahoti, Y.,** Harris, A., Pasquale, L. R., Sun, M., Siesky, B. A., Cohen, G., Wood, S. K., Kwon, M., et al. "An Electronic Health Record-Based Predictive Model to Quantify Multifactorial Risk for Primary Open Angle Glaucoma." ARVO 2026 (Accepted).
- Cohen, G., **Lahoti, Y. S**., Potash, S., Gupta, A., Pasquale, L. R., et al. "Automated Deep-Learning FAZ Quantification and Cardiovascular Risk Associations in the AI-READI Cohort." ARVO 2026 (Accepted).
- Wood, S. K., Harris, A., Muncharaz Duran, L., Schnorbus, N., Gurevitz, C., **Lahoti, Y. S.**, et al. "Electrocardiogram-Synchronized Doppler Holography System for Quantifying Cardiac Cycle-Resolved Retinal Blood Flow." ARVO 2026 (Accepted).
- Greenfield, J., Harris, A., **Lahoti, Y.**, Gupta, A., Pasquale, L. R., et al. "Artificial Intelligence Ready and Exploratory Atlas for Diabetes Insights: the association between glaucoma, hypertension, and cognitive performance." ARVO 2026 (Accepted).
- Siesky, B. A., Harris, A., Potash, S., **Lahoti, Y. S.**, Schanzer, N., et al. "Ocular Perfusion Metrics Demonstrate Strong Coupling with Myocardial Strain and Pulmonary Pressure." ARVO 2026 (Accepted).
- Riina, N., Harris, A., Whitley, D., Siesky, B. A., **Lahoti, Y. S**., Potash, S., et al. "Generative Continuous-Time Recurrent Neural Networks for using Irregularly Sampled Blood Pressure Dynamics to Predict Structural Glaucoma Progression." ARVO 2026 (Accepted).
- Muncharaz Duran, L., Greenberg, L. A., Najac, M. J., Haq, A., **Lahoti, Y.,** et al. "Repeatability of Retinal Blood Flow Metrics Measured Using Doppler Holography." ARVO 2026 (Accepted).
- **Lahoti, Y.**, Yu, J., Cho, S., Kim, J. "Automated Scoliosis Classification from EOS Full Body Imaging: A Deep Learning Approach with RadImageNet." NASS 2024.
- **Lahoti, Y**., Yu, J., Cho, S., Kim, J. "Automated Scoliosis Classification from AI-enabled Spine Contouring." CNS 2024.
- Gold, S. L., Blankemeier, L., **Lahoti, Y.**, et al. "Deep Learning Software to Evaluate Body Composition in Inflammatory Bowel Disease." DDW 2024. ePosters.
- **Lahoti, Y.**, Cho, S., Kim, J. "Development of a Deep Learning Algorithm to Automate the Segmentation of Spinal Cord from EOS Radiographic Images." SGIM 2023.
- **Lahoti, Y.**, Yu, J., Cho, S., Kim, J. "Automated Scoliosis Classification through Classical Machine Learning Methods." EOA 2024.

Scholarly Review:

- Reviewed for Journal: Artificial Intelligence in Vision and Ophthalmology (AIVO) - 2025.

Grant Writing and Scientific Proposal Development

- Connecting Heart and Eye Blood Flow Through a Combination of Artificial Intelligence, Doppler Holography, Echocardiography, and Retrobulbar Imaging | ***NYEE Foundation Grant***.
- Clinical Adaptive Optics Imaging Using a Point-of-Care Commercial Adaptive Optics Scanning Light Ophthalmoscope for Assessment of Retinal Health | ***NYEE Foundation Grant***.
- Artificial Intelligence Analysis of Visual Field Progression Archetypes and Metabolic Biomarkers | ***NYEE Foundation Grant***.
- Integrating Artificial Intelligence and Hemodynamics to Individually Tailor Therapies for Glaucoma | ***BrightFocus Foundation Grant***.
- A Physiology-Enhanced Multimodal Machine Learning Analysis for Predicting Glaucoma Progression in a Large Multicenter Data Consortium | ***National Institutes of Health (NIH) Grant***.

Podium Presentations:

- **Lahoti, Y.**, Yu, J., Cho, S., Kim, J. "Automated Scoliosis Classification from AI-enabled Spine Contouring and Cobb Angle Estimation." EOA 2024.
- **Lahoti, Y.**, Yu, J., Cho, S., Kim, J. "Using Dynamic Time Warping to Find Patterns in Coronal Spinal Alignment." EOA 2024.
- Mohamed, K., Duey, A., Yu, A., **Lahoti, Y.**, et al. "Global Sagittal Alignment Variations with Body Mass Index in Patients." AAOS 2025.

AWARDS

- Founding Member – Society for AI in Vision and Ophthalmology (2025) | Education and Training Committee
- AI Panelist – Invited Faculty: 6th Prostate Cancer Symposium & World Congress of Urologic Oncology ('24)
- NSF National I-Corps Grant – ***SpineSight*** selected from 100+ teams | *NSF Entrepreneurial Training Program* - **$50K Grant**

ADDITIONAL INFORMATION

- Programming: Python, Pytorch, Langchain, Django, React, R, MATLAB, SQL, C++, SolidWorks, Altium, Arduino
- Developed Technologies: Agentic AI, LLM/RAG, Computer Vision, Embedded Machine Learning, Time Series Classification
- Interests: Central Park Cyclist, Heat/Cold Shock Therapy, Mindfulness/Meditation Practitioner, Masala Chai Enthusiast

REFERENCES

Dr. Alon Harris, MS, PhD, FARVO, Director of the Ophthalmic Vascular Diagnostic and Research Program
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Dr. Andrew Deonarine, MD, PhD, Senior Principal Physician Informaticist
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